

## MST-Technologies and Processes

August, 13<sup>th</sup>, 2023

Duration: 120 min.

### Please note

- write your name and matriculation number into the caption of every sheet
- use and write only on the front of the sheet
- use a separate sheet for every task
- write with a pen NOT with a pencil
- if you hinder the exam you will be removed from the lecture hall and fail the exam (5.0 with the statement "Störung")

### Cheating

- any attempt of cheating results in immediate failure of the exam (5.0 with the statement "Täuschung")
- switch off your cell phones and store them in your backpack – a cell phone on the table will be treated as an attempt to cheating

### Withdrawal or absence

- you may withdraw until the moment the tasks are revealed
  - in case you feel unwell during the exam, you may withdraw during the exam for medical reasons; you need to provide (within 3 days) a written note from your doctor confirming the withdrawal for medical reasons
  - please see:  
<https://www.tf.uni-freiburg.de/en/studies-and-teaching/a-to-z-study-faq/de-registration-of-exams>
  - you find the relevant forms here:  
<https://www.tf.uni-freiburg.de/de/studium-lehre/a-bis-z-studium/dokumente/WithdrawalfromExamsduetoIllnessMasterofScience.pdf>
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**Necessary Points:**                      **50 of 100 in total**

**Points:** \_\_\_\_\_

**Result:** \_\_\_\_\_

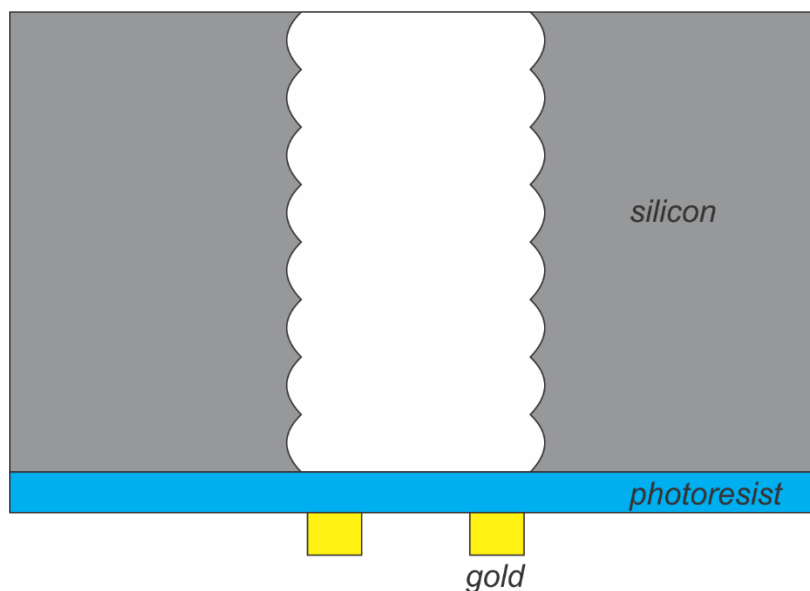


## 1. Processes

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Sketch the manufacturing steps required to fabricate the following structure. The device is deep microchannel, sealed with a layer of a photoresist onto which two gold electrodes are manufacture. These are used as an anemometric flow sensor, i.e., a device which measures the flow velocity in the channel.

Sketch and briefly explain each step. Each manufacturing process (e.g., lithography) used should be depicted out entirely when you first use it. If you use the same process again, you do not need to redraw the entire process.



|                                                                                                        |           |
|--------------------------------------------------------------------------------------------------------|-----------|
| <b>2. Cleanroom technology</b>                                                                         | <b>20</b> |
| 2.1 Name six general recommendations for working in the cleanroom                                      | 6         |
| 2.2 What are common ISO classes used for white areas in a cleanroom for MEMS and for microelectronics? | 4         |
| 2.3 What are the two common cleanroom setups? State the name.                                          | 2         |
| 2.4 What are the four filtering concepts used in filter-fan units? Briefly explain each one            | 8         |

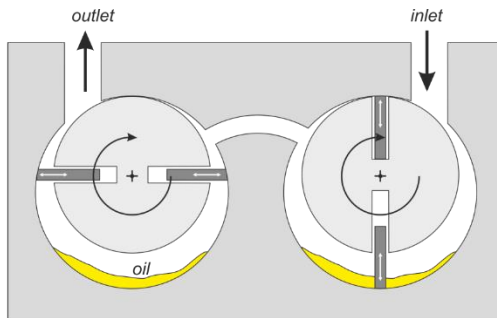


### 3. Vacuum technology

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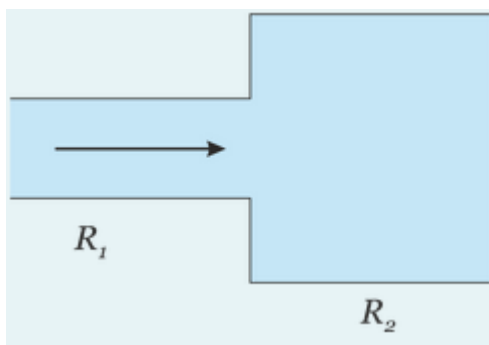
3.1 Which type of vacuum pump is depicted here?

2



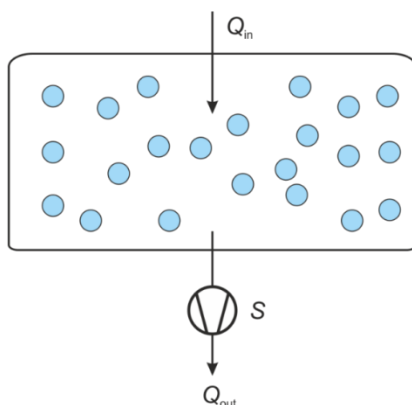
3.2 A hydraulic element with the resistance  $R_1 = 9.5 \text{ s/m}^3$  is connected in series to a second hydraulic element with the resistance  $R_2 = 5.5 \text{ s/m}^3$ . What is the overall hydraulic resistance  $R_{tot}$  of the system?

4



3.3 The following schematic depicts a vacuum chambers which is to be kept at  $p = 0.05 \text{ mbar}$ . The pumping speed of the vacuum pump is  $S = 200 \text{ L/s}$ . What is the maximum allowed gas load  $Q_{in}$  in  $\text{mbar} \cdot \text{L/s}$  which the pump can cope with if the pressure is to be sustained?

4



3.4 Sketch and briefly explain the turbomolecular pump.

5

3.5 Why does the turbomolecular pump always require a backing pump?

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#### 4. Thermal oxidation of silicon

12

4.1 In the Deal-Grove model, the following variables are used:

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- $D$  as the diffusivity of the oxidant in silicon dioxide
- $k$  as the reaction rate of the oxidation of silicon to silicon dioxide
- $N$  as the number of oxidant molecules required to produce one unit volume of oxide
- $C$  as the ambient concentration of the oxidant
- $A = \frac{2D}{k}$  as a simplification factor
- $B = \frac{2DC}{N}$  as a second simplification factor

State the equation for the layer thickness  $x(t)$  at higher layer thicknesses, i.e., for  $x(t > t_c)$ .

4.2 For usage in the Deal-Grove model we determined the following factors:

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- $A = 2 \mu\text{m}$
- $B = 0.25 \mu\text{m}^2/\text{h}$

What is the characteristic time  $t_c$ ? [You do not need a calculator for this - if you find odd numbers, just write down the equation with the numbers]

4.3 Using the same parameters as above, we now run an oxidation process for 4 hours (which is longer than  $t_c$ ). What is the expected layer thickness we achieve? [You do not need a calculator for this - if you find odd numbers, just write down the equation with the numbers]

4

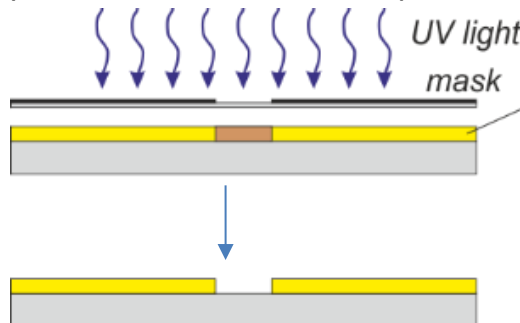


## 5. Photolithography

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- 5.1 The following image depicts two steps in a lithography workflow: the exposure (top) and the development step (below). Which type of photoresist was used here (what should the missing label read)?

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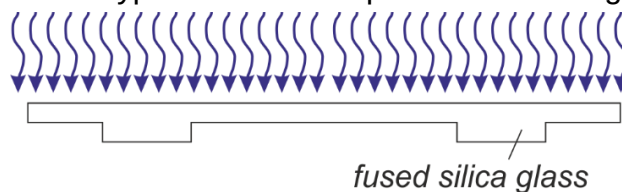


- 5.2 What would be a typical photoresist (brand name, photochemistry on which it is based, etc.) used for the setup depicted in 5.1?
- 5.3 What would you use as developer solution for the photoresist in the setup depicted in 5.1 and the type of photoresist you chose in 5.2?
- 5.4 Which type of mask is depicted in this image?

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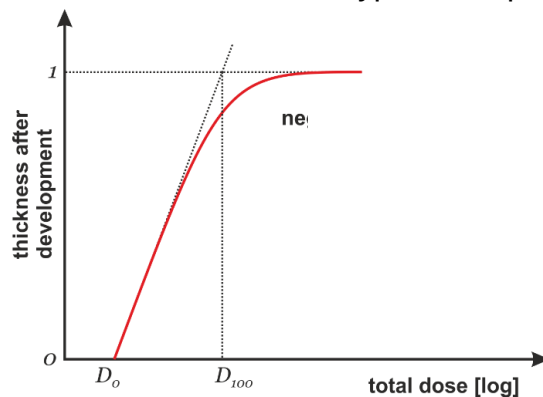


- 5.5 What are the two parameters that a mask can modulate on the incoming light?
- 5.6 Sketch the lithographic setup in proximity lithography. Make sure to include all relevant labels in your sketch.
- 5.7 During characterization of an old bottle of photoresist via an exposure study, you obtain the following graph. Label the characteristic  $D_0$  and  $D_{100}$  values and state the type of the photoresist.

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- 5.8 When choosing the substrate material of a photomask you have several options. What are the most common types of glasses and which one would you chose if you plan to run a lithography at a wavelength of 200 nm? 4
- 5.9 Sketch the classical lithography workflow. Make sure to include all relevant labels. 4

